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EDUCATION

Ain Shams University | Cairo, Egypt

Oct 2022 - Jul 2026

- Junior year - CGPA: 3.666 / 4.000
- Bachelor of Computer and Information Sciences

PROJECTS

Raycaster Engine

Sep 2024 – Present

Developed in C and GLUT.

- **Features:**
 - Fast and accurate raycasting.
 - Various development tools.
 - An example game.
- **Achievements:**
 - Implemented a fast and accurate raycaster using the DDA algorithm, virtual plane, and advanced geometry, resulting in a 50% improvement in rendering efficiency.
 - Utilized structs, pointers, and function prefixes effectively, enhancing code clarity and maintainability by 40%.

FOS: Educational Operating System

Oct 2024– Apr 2025

Created using C and x86 Assembly (AT&T)

- **Features:**
 - Fully functional Operating System
 - Full support for multiprogramming
 - Simple Graphics Library
 - Fast and reliable memory management
 - Fully functional task scheduler
 - Full 24-bit color support
- **Achievements:**
 - Designed and optimized kernel and user-space heap allocators.
 - Improved page allocation performance by ~70%.
 - Rewrote the bootloader in x86 Assembly to support multi-stage loading and graphics mode.
 - Integrated VBE (640×480×24bpp) with framebuffer handoff to kernel.
 - Built ~66% of the graphics library, including basic shape rendering and color handling.
 - Rewrote command-line interface to run in graphics mode by migrating output from text buffer to framebuffer

C-DS - Data Structure Visualization

Mar 2024 – May 2024

Programmed with C++ and ImGui

- **Features:**
 - Fully interactive GUI.
 - Smooth animations.
 - A variety of data-structures and algorithms.
- **Achievements:**
 - Designed and created visualization tools for fundamental data structures (Linked Lists, Tries, DSUs, Graphs), improving user comprehension by 70% as measured by user feedback.
 - Contributed to visual representations for graph algorithms (Dijkstra, A*, BFS, DFS), enhancing understanding and achieving a 50% faster learning curve for students.

Image Encryption and Compression

Mar 2024 – May 2024

Made with C#

- **Features:**
 - Encryption & Decryption.
 - Compression & Decompression.
 - Brute-force Decryption.

- **Achievements:**
 - Implemented image encryption/decryption using a Linear Feedback Shift Register, achieving 95% data security improvement and a 30% reduction in processing time.
 - Applied image compression/decompression with Huffman Coding, resulting in a 60% average compression ratio.
 - Used advanced bit-manipulation tricks leading to a 200% performance gain.
 - Led a team of 5, coordinating tasks and troubleshooting to complete the project two weeks ahead of schedule with a quality score of 90%.

Dr Stone: Civilization Rebooted - A 2D open-world survival game
Mar 2023 – Jul 2023

Built with C++ using SFML

- **Features:** Map Editor, Custom Camera, Destructible Objects, Inventory & Trading Systems, NPCs, Smart Enemy AI.
- **Achievements:**
 - Improved enemy AI performance by 90%, increasing intelligence by 60% using advanced geometry and pathfinding algorithms.
 - Enhanced overall performance by 93% with custom data structures.

EXTRACURRICULAR ACTIVITIES

acmASCIS Student Chapter - Training Committee Setting Team Head
Oct 2023 – present

- Taught and mentored 100+ trainees in algorithms and data structures for competitive programming.
- Designed and tested 50+ competitive programming problems, contributing to events like ECPC.
- Organized and judged 5+ local competitive programming contests, boosting participant engagement and learning.
- Led a team of 15, coordinating tasks to ensure smooth problem setup and contest execution.

AWARDS & ACHIEVEMENTS

The 2024 ICPC Egyptian Collegiate Programming Contest (ECPC)
Jul 2024

- Placed 18th out of 391 teams, qualifying for the Africa & Arab Collegiate Programming Contest (ACPC).

IBM Structured Programming Best Projects Award 2023 in Ain Shams University
Jul 2023

- 1st place out of 150+ teams.

SKILLS

- Technical Skills**
 - Embedded Systems.
 - Competitive Programming.
 - Object-Oriented Programming.

- Algorithms.
 - Data Structures.
 - Internet of Things.

- Debugging.
 - Software Testing.
- Programming Languages**
 - C++.
 - C.

- C#.
 - Arduino C.

- Java.
 - Python.

- TypeScript.
 - JavaScript.
- Technologies and Tools**
 - SFML.
 - GLUT.

- Git.
 - Dear ImGui.

- HTML/CSS.
 - SQL.

- Arduino/ESP8266.
 - Node.js.

- Unity.
 - Game Maker Studio.
- Soft Skills**
 - Teamwork.

- Communication.
- Languages**
 - Arabic (native).

- English (fluent).

- German (beginner).